

# SOFTWARE REQUIREMENTS SPECIFICATION

## PROJECT TITLE: "COLLEGE EVENT MANAGEMENT SYSTEM"

### 1. INTRODUCTION

#### 1.1 Abstract

The College Event Management System is a web-based application developed to manage college events digitally and efficiently. The system allows administrators, faculty, and students to create, organize, register, and monitor events through an online platform. It simplifies event handling, reduces manual work, and improves communication among users.

#### 1.2 Purpose

The purpose of the College Event Management System is to develop a web-based application that enables colleges to manage events efficiently and systematically. The system allows administrators, faculty, and students to create, organize, register, monitor, and participate in events through an online platform. It reduces manual work, improves communication, and enhances the overall event management process.

#### 1.3 Scope

This project provides an online solution for:

- Event creation and management
- Student registration for events
- Approval and monitoring of events
- Viewing schedules and announcements
- Generating event reports

The system reduces manual work and improves communication between organizers and students.

#### 1.3 Objectives

- Automate college event management.
- Simplify event registration.
- Store event records securely.
- Improve event organization and tracking.
- Generate reports for administration.

## **2. FUNCTIONAL REQUIREMENTS**

### **2.1 User Registration and Login**

**Description:**

The system shall allow users to register and log in securely.

**Inputs:**

- Name
- Email ID
- Password
- User Role

**Outputs:**

- Successful registration/login message

### **2.2 Event Creation and Management**

**Description:**

The system shall allow organizers/admin to create, update, and delete event details.

**Inputs:**

- Event Title
- Event Date
- Venue
- Description
- Event Category

**Outputs:**

- Event details stored successfully

### **2.3 Event Registration**

**Description:**

The system shall allow students to register for available events.

**Inputs:**

- Student ID
- Event ID

**Outputs:**

- Registration confirmation

## **2.4 Event Approval**

### **Description:**

The system shall allow admin to approve or reject created events.

### **Inputs:**

- Event request details

### **Outputs:**

- Approval/Rejection status

## **2.5 Attendance Management**

### **Description:**

The system shall maintain student attendance for events.

### **Inputs:**

- Student details
- Attendance status

### **Outputs:**

- Attendance recorded successfully

## **2.6 Notifications and Announcements**

### **Description:**

The system shall send event updates and announcements to users.

### **Inputs:**

- Event notifications
- Announcement message

### **Outputs:**

- Notification displayed to users

## **2.7 Report Generation**

### **Description:**

The system shall generate reports for events and participation.

### **Inputs:**

- Event data
- Attendance data

### **Outputs:**

- Event report
- Participation report

## **2.8 User Profile Management**

### **Description:**

The system shall allow users to update profile details.

### **Inputs:**

- Name
- Contact Information
- Password

### **Outputs:**

- Updated profile information

### **3. NON - FUNCTIONAL REQUIREMENTS**

#### **3.1 Performance**

- Support multiple users simultaneously.
- Response time < 3 seconds.

#### **3.2 Security**

- Password encryption
- Secure login authentication

#### **3.3 Reliability**

- Database backup support

#### **3.4 Usability**

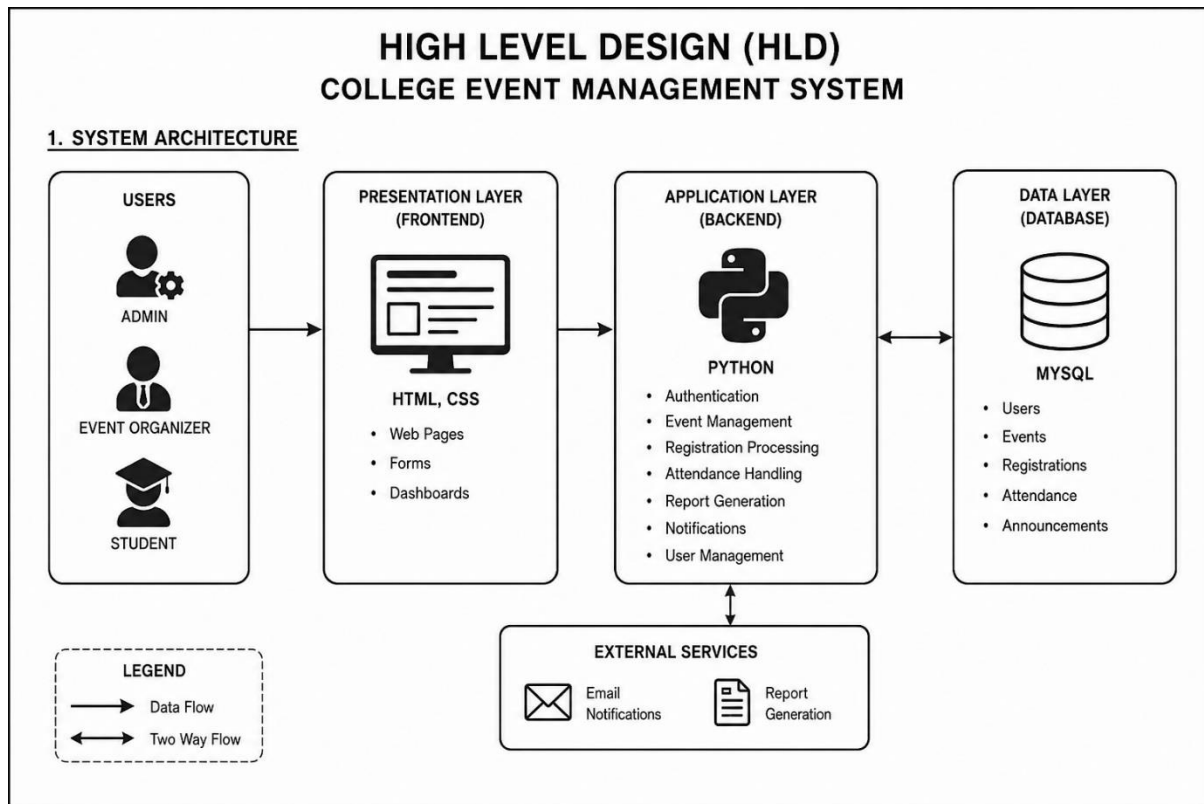
- Easy-to-use interface

#### **3.5 Availability**

- System available 24/7

## 4. HIGH LEVEL DESIGN (HLD)

### System Architecture



### Module Design

#### Module 1: User Authentication Module

Functions:

- User Registration
- Login
- Password Validation
- Role-based Access

Input:

- Email, Password

Output:

- User Dashboard

## **Module 2: Event Management Module**

Functions:

- Create Event
- Update Event
- Delete Event
- View Events

Input:

- Event details

Output:

- Event List

## **Module 3: Event Registration Module**

Functions:

- Register Event
- Cancel Registration
- View Registered Events

Input:

- Student ID
- Event ID

Output:

- Registration Status

## **Module 4: Attendance Module**

Functions:

- Mark Attendance
- View Attendance

Input:

- Student Details

Output:

- Attendance Report

### **Module 5: Notification Module**

Functions:

- Send Announcements
- Event Updates

Output:

- Notification Display

### **Module 6: Report Generation Module**

Functions:

- Generate Reports
- Export Event Data

Output:

- Event Reports

### **Module 7: Admin Dashboard Module**

Functions:

- Manage Users
- Approve Events
- View Statistics

Output:

- Dashboard Analytics



## **5. LOW LEVEL DESIGN (LLD)**

### **1. User Authentication Module**

**Purpose:** Manage user registration and login.

**Inputs:**

- Name
- Email
- Password
- Role

**Process:**

- Validate user input
- Check email uniqueness
- Encrypt password
- Store details in database
- Authenticate login credentials

**Outputs:**

- Login success/failure
- User dashboard

**Database Table:** Users

### **2. Event Management Module**

**Purpose:** Create and manage events.

**Inputs:**

- Event Title
- Date
- Venue
- Description
- Category

**Process:**

- Validate event details
- Save event data
- Update/Delete event information

**Outputs:**

- Event created/updated message

**Database Table:** Events

### 3. Event Registration Module

**Purpose:** Allow students to register for events.

**Inputs:**

- Student ID
- Event ID

**Process:**

- Verify event availability
- Check duplicate registration
- Store registration data

**Outputs:**

- Registration confirmation

**Database Table:** Registration

### 4. Attendance Management Module

**Purpose:** Track event attendance.

**Inputs:**

- Student ID
- Attendance Status

**Process:**

- Retrieve registered students
- Mark attendance
- Save attendance records

**Outputs:**

- Attendance status

**Database Table:** Attendance

## 5. Notification Module

**Purpose:** Send event updates.

**Inputs:**

- Event Announcement
- Notification Content

**Process:**

- Generate notification
- Send notification to users

**Outputs:**

- Notification delivered

**Database Table:** Notifications

## 6. Report Generation Module

**Purpose:** Generate event reports.

**Inputs:**

- Event Data
- Attendance Data

**Process:**

- Collect required records
- Generate summary report

**Outputs:**

- Event Report
- Attendance Report

**Database Table:** Events, Attendance

## 7. Admin Module

**Purpose:** Manage system operations.

**Inputs:**

- User Information
- Event Requests

**Process:**

- Approve events
- Manage users
- View reports

**Outputs:**

- Admin dashboard updates

**Database Tables:** Users, Events

## **6. SYSTEM REQUIREMENTS**

### **6.1 Hardware Requirements**

- Processor: Intel Core i3 or above
- RAM: Minimum 4 GB
- Storage: Minimum 100 GB
- Monitor: 1366 × 768 resolution or higher
- Keyboard and Mouse

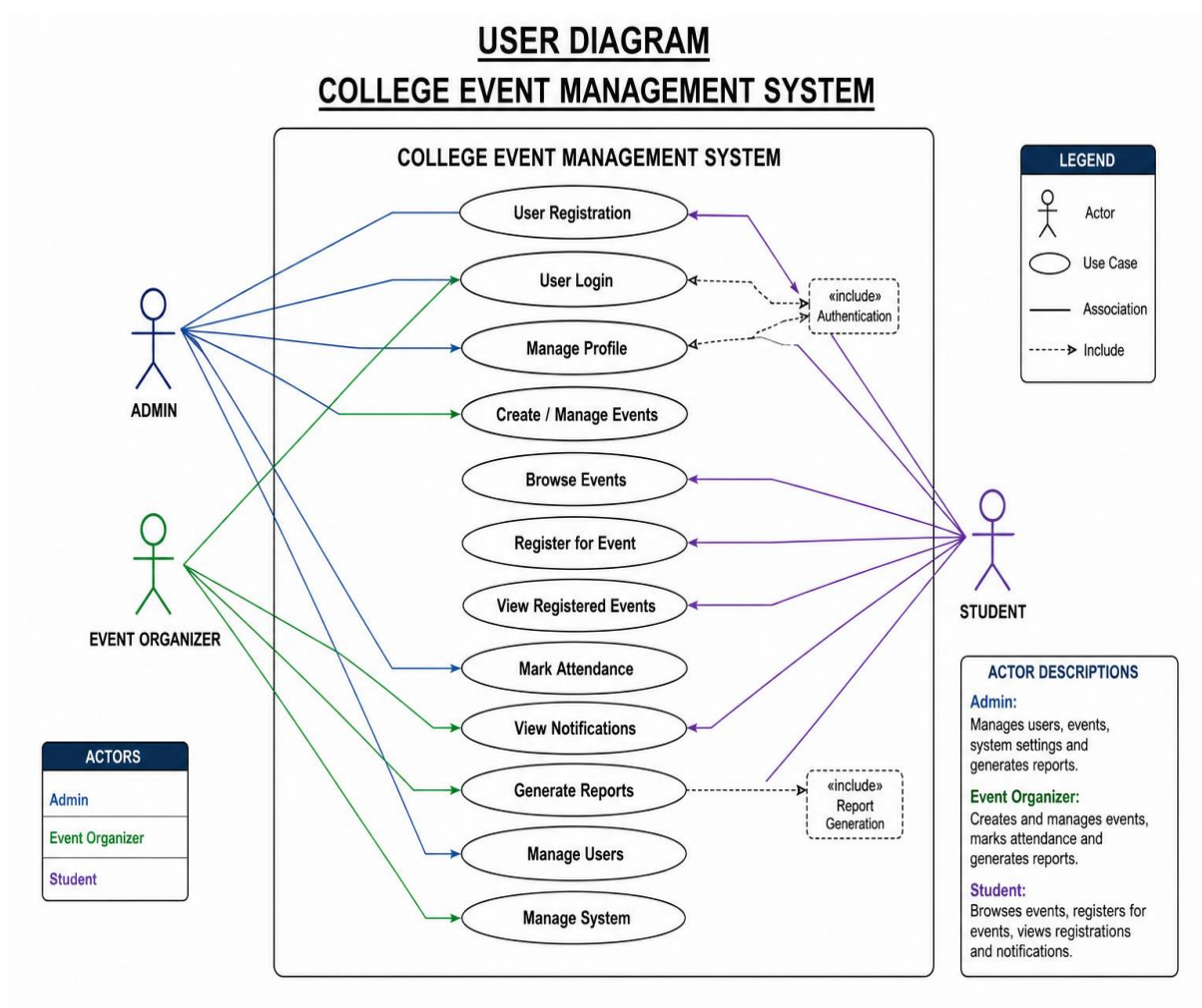
### **6.2 Software Requirements**

- Operating System: Windows 10/11 or Linux
- Frontend: HTML, CSS
- Backend: Python
- Database: MySQL
- IDE/Editor: VS Code
- Web Browser: Google Chrome, Microsoft Edge, Firefox

## 7. TECH STACK

| LAYER            | TECHNOLOGY                     |
|------------------|--------------------------------|
| Frontend         | HTML, CSS                      |
| Backend          | Python                         |
| Database         | MySQL                          |
| Development Tool | VS Code                        |
| Browser          | Google Chrome / Microsoft Edge |

## 8. USER DIAGRAM



## **9. FUTURE ENHANCEMENTS**

- 1. Mobile Application Support**
  - Access and manage events using mobile devices.
- 2. QR Code Attendance**
  - Mark attendance quickly using QR codes.
- 3. Email Notifications**
  - Send event updates and reminders automatically.
- 4. Certificate Generation**
  - Generate participation certificates automatically.
- 5. Event Analytics Dashboard**
  - View event statistics and reports.
- 6. Online Registration and Payment**
  - Allow students to register and pay online for events.

## **10. CONCLUSION**

The College Event Management System is developed to simplify and automate the process of organizing and managing college events. The system enables administrators, organizers, and students to create events, register participants, manage attendance, and generate reports efficiently. By using HTML, CSS, Python, and MySQL, the project reduces manual work, improves communication, and provides a secure and user-friendly platform for event management. This system can be further enhanced with additional features in future versions.